

Smart Fleet Selection: Optimisation for Safety, Sustainability, and Cost

Maritime Hackathon 2026

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1. Executive Summary

1.1 Tri-Objective Optimisation

This project addresses a Tri-Objective Optimisation challenge: selecting an optimal fleet to transport bunker fuel from Singapore to Australia's West Coast within one month while balancing three competing goals — minimising costs (amortised and operational), maintaining fleet-wide average safety scores ≥ 3.0 , and reducing CO₂eq emissions. Total number of vessels chosen is 22.

1.2 Cargo Demand Calculation

Using the MPA 2024 Annual Report (Page 10), total annual bunker sales across all fuel types reached 54.92 million tonnes. Dividing by 12 yields a monthly target demand (D) of 4.5767 million tonnes. The selected fleet's combined Deadweight Tonnage (DWT) must meet or exceed this capacity.

1.3 Methodology

The selection engine employs an Integrated Cost Model incorporating four factors: amortised ownership, fuel consumption, carbon taxation, and safety-based risk adjustments.

- **Energy-Based Fuel Logic:** Fuel is priced by energy (USD/GJ) and converted per machinery using its LCV (ME/AE/Boiler).
- **Emission Quantification:** CO₂e sums CO₂, CH₄, N₂O from all machinery during transit/manoeuvre using GWPs 1, 28, 265.

Fleet selection is formulated as a binary cost-minimisation for the one-way route Singapore → Port Hedland, meeting monthly demand D. Candidate vessels must show Singapore port-boundary followed by Port Hedland in AIS, and only transit + maneuvering time is costed. Engine load is estimated using $LF = (AS/MS)^3$ (with a minimum movement floor), and fuel/emissions are computed separately for ME/AE/Boiler using their respective fuel types.

Total monthly vessel cost is fuel + carbon + amortised ownership, then scaled by a safety risk premium. Fuel cost follows the clarification: cost per tonne = USD/GJ $\times$ LCV for each machinery fuel type, summed as ME + AE + Boiler. Emissions include LLAF at low load before conversion to CO₂eq and carbon pricing. The optimal fleet minimises $\sum \text{Cost}_i x_i$ subject to $\sum \text{DWT}_i x_i \geq D$, average safety $\sum (s_i - S_{\min}) x_i \geq 0$, and ≥ 1 vessel per main fuel type, with $x_i \in \{0,1\}$. Baseline uses $S_{\min}=3$; sensitivity uses $S_{\min}=4$.

2. Constraints & Fleet Diversity Analysis

2.1 Diversity and Safety Requirements

To ensure long-term industry resilience, the fleet selection process is governed by specific operational "floors" that maintain both technological diversity and safety standards.

The first primary constraint is a fuel mix strategy, which mandates the inclusion of at least one vessel from every major Main Engine fuel category, such as Distillate, LNG, and Methanol. This requirement ensures the model can effectively evaluate and compare performance across various propulsion technologies, rather than relying overly on a single energy source.

Furthermore, the fleet must adhere to a safety floor based on a standardised scoring system ranging from 1 (high risk) to 5 (low risk). To maintain overall integrity, the fleet's average safety score must remain at or above 3.0. This is managed through a "Cost of Risk" selection logic, which applies financial premiums as penalties to higher-risk vessels while offering negative premiums, or rewards, to those with superior safety ratings.

3. Sensitivity Outcome Stretch Objective Analysis

3.1 Sensitivity Analysis: Enhanced Safety Requirement (Score ≥ 4.0)

Increasing the safety constraint from ≥ 3.0 to ≥ 4.0 raised total cost 7.81% (\$1.63M) while expanding fleet size from 22 to 23 vessels. To achieve a fleet average safety score ≥ 4.0 , we replaced 6 high-risk Distillate vessels and 3 mid-tier alternative fuel ships with 10 high-safety

modern vessels, including 3 LNG, 1 Ammonia, 1 LPG, and 5 other alternative fuel ships. Emissions increased only 2.20% despite 12.19% higher fuel consumption—a 5.5× differential driven by fleet shifts toward cleaner alternative fuels (LNG, Ammonia) with superior carbon intensity but lower operational efficiency.

Metric	Baseline Fleet (Average ≥ 3.0)	Safety-Pivot (Average ≥ 4.0)	Variance
Total Cost (USD)	\$20,894,362.60	\$22,525,772.25	+7.81%
Fleet Size (Ships)	22	23	+1
Average Safety Score	3.05	4.00	+0.95
Total CO2e (Tonnes)	14890.374	15217.843	+2.20%
Total Fuel (Tonnes)	5134.040	5759.807	+12.19%

3.2 Cost-Safety Trade-off Analysis

The analysis reveals the significant premium for enhanced safety. To meet the higher 4.0 safety target while maintaining cargo volume, the model had replaced cheaper, high-capacity vessels with safer alternatives that were more expensive or less efficient. This substitution directly caused the 7.81% cost increase. Crucially, the 12.19% surge in fuel use, far outpacing the 2.20% rise in emissions, indicates the available safer vessels were predominantly less fuel-efficient, though some likely used cleaner fuels with lower emission factors.

3.3 Conclusions and Recommendations

Under current vessel economics, stricter safety mandates force a trade-off between cost, fuel use, and emissions. Fleet planners must therefore prioritize multi-criteria screening to identify vessels that balance safety with efficiency. Where possible, operational measures like speed optimization should be evaluated to achieve risk reduction without costly fleet upgrades. Ultimately, the "optimal" fleet depends on accurately valuing safety against financial and environmental performance.